

SECTION A (25 marks)

This section consists of **3** questions. Answer **all** questions.

1. Evaluate the following limits, if exists:

(a) $\lim_{x \rightarrow -\infty} \frac{3x}{\sqrt{x^2 - 36}}$. [4 marks]

(b) $\lim_{x \rightarrow 3} \frac{\sqrt{x^2 + 7} - 4}{x^2 - 3x}$. [5 marks]

2. (a) Find the derivative of $f(x) = \sqrt{8x - 3}$ by using the first principle of differentiation. [4 marks]

(b) Find $\frac{dy}{dx}$ for the following:

(i) $y = \sin(\ln 3x)$ [3 marks]

(ii) $y = \frac{x}{x^3 + 1} + e^{2x} \tan x$. [4 marks]

3. Find the inflection point of the function $f(x) = x^3 + 3x^2 - 24x + 20$. [5 marks]

SECTION B (75 marks)

This section consists of 7 questions. Answer **all** questions.

1. Given $z_1 = 3 + 4i$ and $z_2 = 5 - 2i$. Find z_3 in $a + bi$ form if $\frac{1}{z_3} = \frac{1}{z_1} + \frac{1}{z_1 z_2}$.

Hence, express $\overline{z_3}$ in polar form.

[9 marks]

2. (a) Solve the equality $|3x + 1| = |2x - 7|$.

[4 marks]

- (b) Obtain solution set for the following inequalities

(i) $|2x + 7| + 1 \geq 6x$.

[4 marks]

(ii) $\left| \frac{2x+1}{x-3} \right| \geq 2$.

[4 marks]

3. (a) A function $f(x)$ is defined as

$$f(x) = \frac{3 - \ln(4-x)}{2}, \quad x \in \mathbb{R} \text{ and } x < 4.$$

- (i) Determine the inverse function, $f^{-1}(x)$.

[3 marks]

- (ii) Determine whether $f^{-1}(x)$ is one to one.

[3 marks]

- (b) Given the function $g(x) = e^{x+1}$. Find the function $h(x)$ if $(h \circ g)(x) = x + 1$.

[6 marks]

4. A function f is defined as

$$f(x) = -(x-2)^2 + 1, \quad x \geq 2$$

Determine the inverse function of f .

Hence, sketch the graph of f and f^{-1} on the same axes.

[6 marks]

5. (a) Find the vertical and horizontal asymptotes of

$$f(x) = \frac{2x^2 + 1}{4x^2 - 25}$$

[6 marks]

(b) Given

$$f(x) = \begin{cases} \frac{x^2 - 16}{x^2 - 4x}, & x \neq 4 \\ p & x = 4 \end{cases}$$

where p is a constant.

(i) Determine $\lim_{x \rightarrow 4} f(x)$ if exists. [3 marks]

(ii) Find the value of p if $f(x)$ is continuous at $x = 4$. [2 marks]

6. (a) The parametric equations of a curve is given by $x = -9\sin^3 t$ and $y = 3 - 3\cos^3 t$.

(i) Show that $\frac{dy}{dx} = -\frac{1}{3}\cot t$. Hence, solve $\frac{dy}{dx} = 0$ for $0 \leq t \leq \pi$. [8 marks]

(ii) Show that $\frac{d^2y}{dx^2} = -\frac{1}{81\sin^4 t \cdot \cos t}$. [4 marks]

(b) Given $e^{-2y} + 2xy + \ln(1 - 2x) = 2$. Calculate the value of $\frac{dy}{dx}$ when $x = 2$ and $y = 0$. [4 marks]

7. (a) Given a water tank which has the shape of an inverted circular cone water tank with a base radius of 3 meters and a height of 5 meters. If water is being pumped into the tank at the rate of $2 \text{ m}^3/\text{minute}$, find the rate of change in water level rising when the water is 3 meters deep. [5 marks]

(b) Air is pumped into a spherical balloon at the rate of $12\pi \text{ cm}^3 \text{ s}^{-1}$. Find the rate of increase of the surface area when its radius is 6 cm. [4 marks]

END OF QUESTION

Final Answer:

SECTION A

1. (a) -3 (b) $\frac{1}{4}$
2. (a) $\frac{4}{\sqrt{8x-3}}$ (b)(i) $\frac{\cos(\ln 3x)}{x}$ (b)(ii) $\frac{1-2x^3}{(x^3+1)^2} + e^{2x}(\sec^2 x + 2 \tan x)$
3. Point of inflection = (-1, 46)

SECTION B

1. $z_3 = \frac{11}{4} + \frac{13}{4}i$; $\overline{z_3} = 4.26(\cos(-0.8685) + i \sin(-0.8685))$
2. (a) $x = \frac{6}{5}$, $x = -8$
(b)(i) $\{x: x \leq 2\}$ (ii) $\{x: x \geq \frac{5}{4}, x \neq 3\}$
3. (a)(i) $f^{-1}(x) = 4 - e^{3-2x}$ (ii) $f^{-1}(x)$ is one to one
(b) $h(x) = \ln x$
4. $f^{-1}(x) = 2 + \sqrt{1-x}$, $x \leq 1$, D.I.Y
5. (a) V.A: $x = \frac{5}{2}$, $x = -\frac{5}{2}$, H.A: $y = \frac{1}{2}$ (b) (i) $\lim_{x \rightarrow 4} f(x) = 2$ (ii) $p = 2$
6. (a) (i) $\frac{\pi}{2}$ (ii) D.I.Y (b) $\frac{dy}{dx} = -\frac{1}{3}$
7. (a) $\frac{50}{81\pi} \text{ cm s}^{-1}$ (b) $4\pi \text{ cm}^2 \text{ s}^{-1}$